

Drugs acting on the respiratory system

Respiratory system

- ❖ The respiratory system includes
 - the upper airway passages,
 - the nasal cavities,
 - pharynx and trachea as well as the
 - bronchi and bronchioles.
- ❖ Respiration is the exchange of gases between the **tissue** of the body and to **outside environment**.
- ❖ It involves breathing in of **an air** through **the respiratory tract, uptake of oxygen from the lungs, transport of oxygen through the body in the blood stream, utilization of oxygen in the metabolic activities** (cells and removal of carbon dioxide from the body)

Disorders of Respiratory Function

Main disorders of the respiratory system are

- 1. Bronchial asthma**
- 2. Cough**
- 3. Allergic rhinitis**
- 4. Chronic obstructive pulmonary disease (COPD, also called emphysema)**

Asthma

Asthma is a chronic inflammatory disorder of bronchial airways that result in airway obstruction in response to external stimuli (as pollen grains, cold air and tobacco smoke).

studies.

Characters of airways in asthmatic patients :

Airway hyper-reactivity: abnormal sensitivity of the airways to wide range of external stimuli.

Inflammation

- Swelling
- Thick mucus production.

Bronchospasm (constriction of the bronchial muscles).

Symptoms of asthma

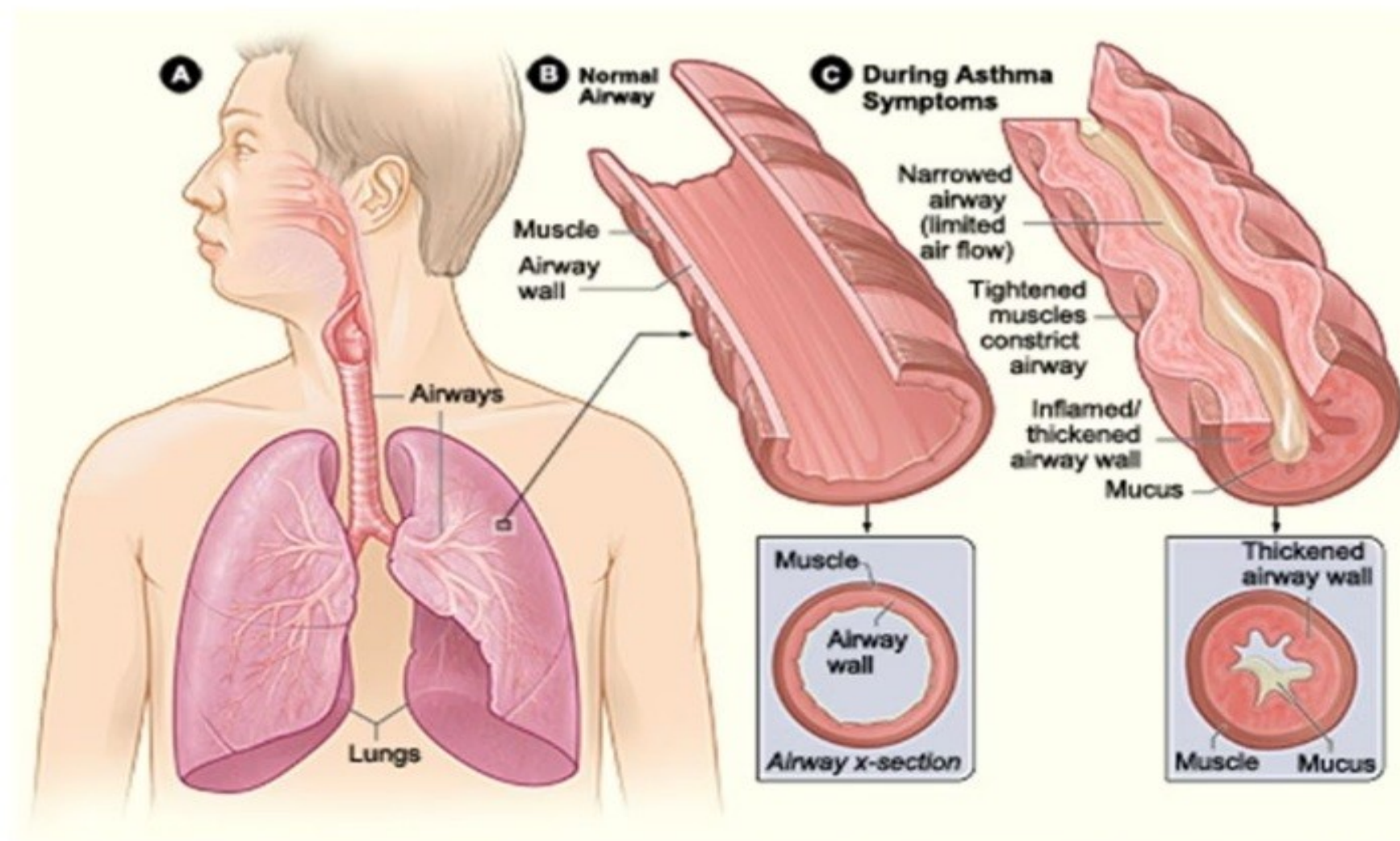
Asthma produces recurrent episodic attack of

- **Acute bronchoconstriction**
- **Shortness of breath**
- **Chest tightness**
- **Wheezing**
- **Rapid respiration**
- **Cough**

Symptoms can happen each time the airways are irritated by inhaled irritants or allergens.

Causes

- **Infection**
- **Emotional conditions**
- **Stress**
- **Exercise**
- **Pets**
- **Seasonal changes**
- **Some drugs as aspirin, β blockers**



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Classifications

		Persistent			
	Components	Intermittent	Mild	Moderate	Severe
Impairment	Symptoms	≤2 days/wk	>2 days/wk but not daily	Daily	Throughout the day
	Nighttime awakenings	≤2 × month	3–4 × month	>1 × week, but not nightly	Often 7 × week
	SABA use for Sx control	≤2 days/wk	>2 days/wk, but not >1 × day	Daily	Several times per day
	Interference with normal activity	None	Minor limitation	Some limitation	Extremely limited
	Lung function ^a	FEV ₁ >80% FEV ₁ /FVC normal	FEV ₁ >80% FEV ₁ /FVC normal	FEV ₁ 60–80% FEV ₁ /FVC reduced 5% (0.05)	FEV ₁ <60% FEV ₁ /FVC reduced >5% (>0.05)
		Intermittent	Persistent		
Risk	Exacerbations	0–2/y	>2 in 1 year	—————→	
	Recommended step for initiating treatment	Step 1	Step 2	Step 3 and consider short course of oral corticosteroids	Step 4 or 5 course of oral corticosteroid

Pathophysiology

- ❖ Inhaled allergen challenge in allergic patients leads to an **early phase reaction**

Mediated by immunoglobulin E (IgE) that,

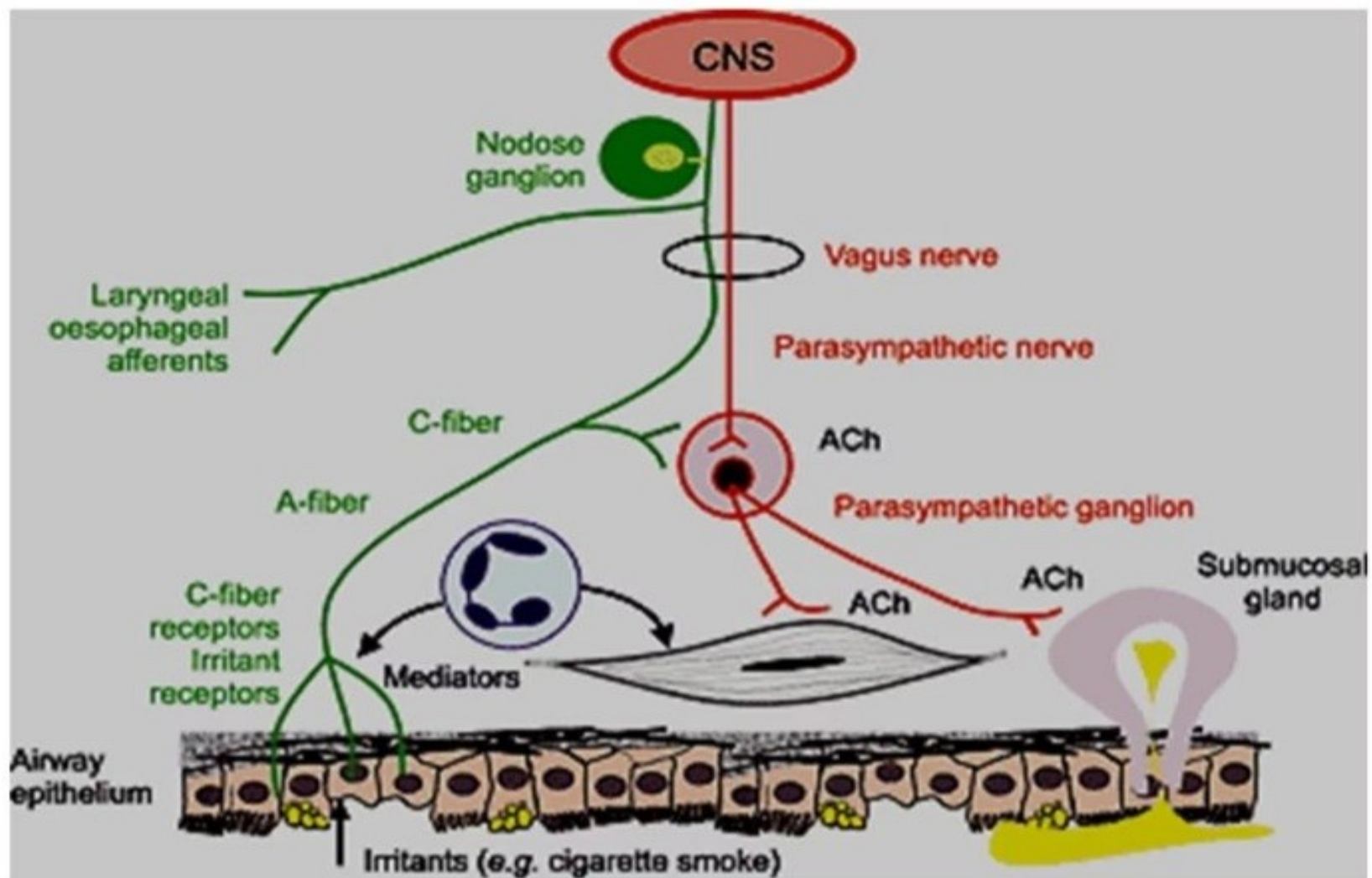
- ❖ Rapid activation of airway mast cells and macrophages
- ✓ leading to the rapid release of proinflammatory mediators such as
 - histamine,
 - eicosanoids, and
 - reactive oxygen (O₂) species that
 - induce contraction of airway smooth muscle, mucus secretion, and vasodilation.⁷
 - The bronchial microcirculation has an essential role in

Pathophysiology...

- ❖ Inflammatory mediators induce microvascular leakage with exudation of plasma in the airways
 - Acute plasma protein leakage induces a thickened, engorged, and edematous
 - consequent narrowing of the airway lumen

Pathophysiology...

- ❖ The late-phase inflammatory reaction occurs **6 to 9 hours** after allergen provocation
- ❖ Involves the recruitment and activation of eosinophils,
 - CD4+ thymically derived lymphocytes (T cells),
 - basophils,
 - neutrophils, and
 - macrophages.



Aims of anti asthmatic drugs:

- **To relieve acute episodic attacks of asthma (*bronchodilators, quick relief medications*).**
- **To reduce the frequency of attacks, and nocturnal awakenings (*anti-inflammatory drugs, prophylactic or control therapy*).**

Anti asthmatic drugs

Bronchodilators

(Quick relief medications)

treat acute episodic attack of asthma

- **Short acting β 2-agonists**
- **Antimuscarinics**
- **Xanthine preparations**

Anti-inflammatory Agents
(control medications or prophylactic therapy)

reduce the frequency of attacks

- **Corticosteroids**
- **Mast cell stabilizers**
- **Leukotrienes antagonists**
- **Anti-IgE monoclonal antibody**
- **Long acting β 2-agonists**

Anti asthmatic drugs

Bronchodilators : (Quick relief medications)
are used to relieve acute attack of
bronchoconstriction

- 1. β_2 - adrenoreceptor agonists**
- 2. Antimuscarinics**
- 3. Xanthine preparations**

Sympathomimetics

β -adrenoceptor agonists

Mechanism of Action

- **direct β_2 stimulation $\longrightarrow$ stimulate adenylyl cyclase $\longrightarrow$ Increase cAMP $\longrightarrow$ bronchodilation**
- **Inhibit mediators release from mast cells.**
- **Increase mucus clearance by (increasing ciliary activity).**

Classification of β agonists

- **Non selective β agonists:**
epinephrine - isoprenaline
- **Selective β_2 – agonists (Preferable).**
Salbutamol (albuterol)
Terbutaline
Salmeterol
Formeterol

Non selective β -agonists.

Epinephrine

- **Potent bronchodilator**
- **rapid action (maximum effect within 15 min).**
- **S.C. or by inhalation (aerosol or nebulizer).**
- **Has short duration of action (60-90 min)**
- **Drug of choice** for acute anaphylaxis
(hypersensitivity reactions).

Nebulizer



Inhaler



Disadvantages

- **Not effective orally.**
- **Hyperglycemia**
- **CVS side effects:**
 - tachycardia, arrhythmia, hypertension
- **Skeletal muscle tremor**
- **Not suitable for asthmatic patients with hypertension or heart failure.**

Contraindication:

CVS patients, diabetic patients

Selective β_2 –agonists

- drugs of choice for acute attack of asthma
- Are mainly given by **inhalation** (metered dose inhaler or nebulizer).
- Can be given orally, parenterally.
- **Short acting β_2 agonists**
e.g. salbutamol, terbutaline
- **Long acting β_2 agonists**
e.g. salmeterol, formeterol

Short acting β_2 agonists

Salbutamol, inhalation, orally, i.v.

Terbutaline, inhalation, orally, s.c.

- **Have rapid onset of action (15-30 min).**
- **short duration of action (4-6 hr)**
- **used for symptomatic treatment of acute episodic attack of asthma.**

Long acting selective β_2 agonists

- **Salmeterol & formoterol:**
- **Long acting bronchodilators (12 hours)**
- **have high lipid solubility (creates depot effect)**
- **are given by inhalation**
- **are not used to relieve acute episodes of asthma**
- **used for nocturnal asthma (long acting relievers).**
- **combined with inhaled corticosteroids to control asthma (*decreases the number and severity of asthma attacks*).**

Advantages of β_2 agonists

- **Minimal CVS side effects**
- **suitable for asthmatic patients with hypertension or heart failure.**

Disadvantages of β_2 agonists

- **Skeletal muscle tremors.**
- **Nervousness**
- **Tolerance (B-receptors down regulation).**
- **Tachycardia over dose (B1-stimulation).**

Muscarinic antagonists

Ipratropium – Tiotropium

- Act by blocking muscarinic receptors.
- Given by aerosol inhalation
- Quaternary derivatives of atropine
- Does not diffuse into the blood
- Do not enter CNS, minimal systemic side effects.
- Delayed onset of action
- **Ipratropium** has short duration of action 3-5 hr
- **Tiotropium** has longer duration of action (24 h).

Pharmacodynamics

- are short-acting bronchodilator.
- Inhibit bronchoconstriction and mucus secretion
- Less effective than β 2-agonists.
- No anti-inflammatory action

Uses

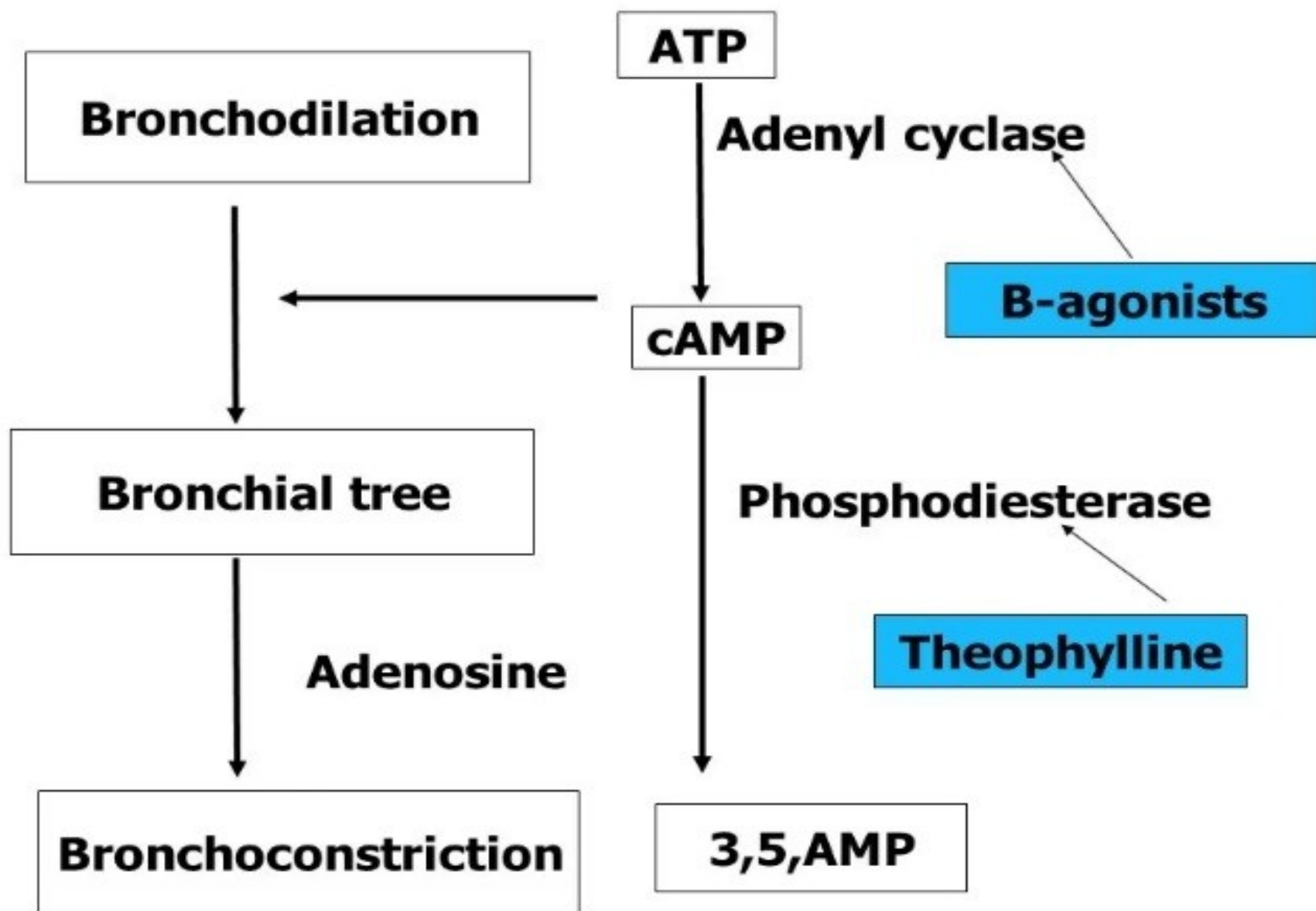
- Main choice in chronic obstructive pulmonary diseases (COPD).
- In acute severe asthma combined with β 2-agonists & steroids.

Methylxanthines

➤ **Theophylline - aminophylline**

Mechanism of Action

- **are phosphodiesterase inhibitors**
- **↑ cAMP → bronchodilation**
- **Adenosine receptors antagonists (A1)**
- **Increase diaphragmatic contraction**
- **Stabilization of mast cell membrane**



Pharmacological effects :

➤ **Bronchial muscle relaxation**

➤ **↑ contraction of diaphragm → improve ventilation**

CVS: ↑ heart rate, ↑ force of contraction

GIT: ↑ gastric acid secretions

Kidney: ↑ renal blood flow, weak diuretic action

CNS stimulation

- * stimulant effect on respiratory center.**
- * decrease fatigue & elevate mood.**
- * overdose (tremors, nervousness, insomnia, convulsion)**

Pharmacokinetics

➤ **metabolized by Cyt P450 enzymes in liver**

➤ $T_{1/2} = 8$ hours

➤ **has many drug interactions**

➤ **Enzyme inducers:** as phenobarbitone-rifampicin $\rightarrow$ $\uparrow$ metabolism of theophylline $\rightarrow$ $\downarrow$ $T_{1/2}$.

➤ **Enzyme inhibitors:** as erythromycin $\rightarrow$ $\downarrow$ metabolism of theophylline $\rightarrow$ $\uparrow T_{1/2}$.

Uses

- **Second line drug in asthma (theophylline)**
- **For status asthmatics (aminophylline, is given as slow infusion).**

Side Effects

- **Low therapeutic index** narrow safety margin *monitoring of theophylline blood level is necessary.*
- **CVS effects:** hypotension, arrhythmia.
- **GIT effects:** nausea & vomiting
- **CNS side effects:** tremors, nervousness, insomnia, convulsion

Anti - inflammatory agents include:

- **Glucocorticoids**
- **Leukotrienes antagonists**
- **Mast cell stabilizers**
- **Anti-IgE monoclonal antibody (omalizumab)**

Anti - inflammatory Agents:

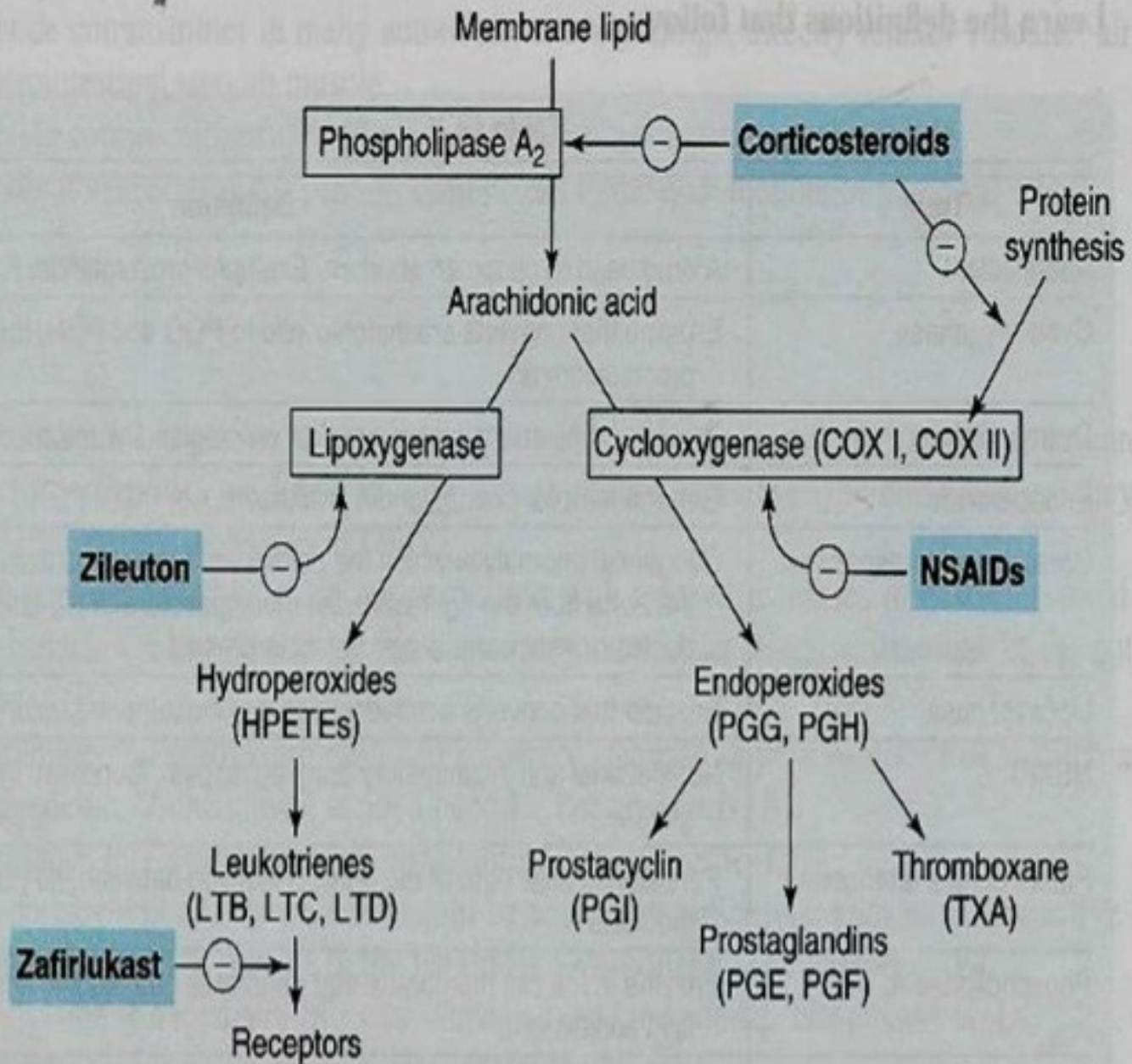
(control medications / prophylactic therapy)

reduce the number of inflammatory cells in the airways and prevent blood vessels from leaking fluid into the airway tissues. By reducing inflammation, they reduce the spasm of airways & bronchial hyper-reactivity.

Glucocorticoids

Mechanism of action

- Inhibition of *phospholipase A2*
- ↓ prostaglandin and leukotrienes
- ↓ Number of inflammatory cells in airways.
- *Mast cell stabilization* → ↓ histamine release.
- ↓ capillary permeability and mucosal edema.
- Inhibition of antigen-antibody reaction.
- **Upregulate β_2 receptors** (have additive effect to B2 agonists).



Pharmacological actions of glucocorticoids

- Anti-inflammatory actions
- Immunosuppressant effects
- Metabolic effects
 - Hyperglycemia
 - ↑ protein catabolism, ↓ protein anabolism
 - Stimulation of **lipolysis** - fat redistribution
- Mineralocorticoid effects:
 - sodium/fluid retention
 - Increase potassium excretion (hypokalemia)
 - Increase blood volume (hypertension)

- Behavioral changes: depression
- Bone loss (osteoporosis) due to
 - Inhibit bone formation
 - ↓ calcium absorption.

Routes of administration

➤ **Inhalation:**

e.g. *Budesonide & Fluticasone, beclometasone*

- Given by inhalation, given by metered-dose inhaler
- Have first pass metabolism
- Best choice in asthma, less side effects

➤ **Orally:** Prednisone, methyl prednisolone

➤ **Injection:** Hydrocortisone, dexamethasone

Glucocorticoids in asthma

- Are **not** bronchodilators
- Reduce bronchial inflammation
- Reduce bronchial hyper-reactivity to stimuli
- Have delayed onset of action (effect usually attained after 2-4 weeks).
- Maximum action at 9-12 months.
- Given as prophylactic medications, used alone or combined with beta-agonists.
- Effective in allergic, exercise, antigen and irritant-induced asthma,

Systemic corticosteroids are reserved for:

- Status asthmaticus (i.v.).

Inhaled steroids should be considered for adults, children with any of the following features

- using inhaled β_2 agonists three times/week
- symptomatic **three times/** week or more;
- or waking **one night/**week.

Clinical Uses of glucocorticoids

1. Treatment of **inflammatory disorders** (asthma, rheumatoid arthritis).
2. Treatment of **autoimmune disorders** (ulcerative colitis, psoriasis) and after organ or bone marrow transplantation.
3. **Antiemetics** in cancer chemotherapy

Side effects due to systemic corticosteroids

- Adrenal suppression
- Growth retardation in children
- Osteoporosis
- Fluid retention, weight gain, hypertension
- Hyperglycemia
- Susceptibility to infections
- Glaucoma
- Cataract
- Fat distribution, wasting of the muscles
- Psychosis

Inhalation has very less side effects:

- Oropharyngeal candidiasis (thrush).
- Dysphonia (voice hoarseness).

Withdrawal

- Abrupt stop of corticosteroids should be avoided and dose should be tapered (*adrenal insufficiency syndrome*).

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Mast cell stabilizers

e.g. Cromolyn (cromoglycate) - Nedocromil

- act by stabilization of mast cell membrane.
- given by inhalation (aerosol, microfine powder, nebulizer).
- Have poor oral absorption (10%)

Pharmacodynamics

- are Not bronchodilators
- Not effective in acute attack of asthma.
- Prophylactic anti-inflammatory drug
- Reduce bronchial hyper-reactivity.
- Effective in exercise, antigen and irritant-induced asthma.
- Children respond better than adults

Uses

- Prophylactic therapy in asthma especially in children.
- Allergic rhinitis.
- Conjunctivitis.

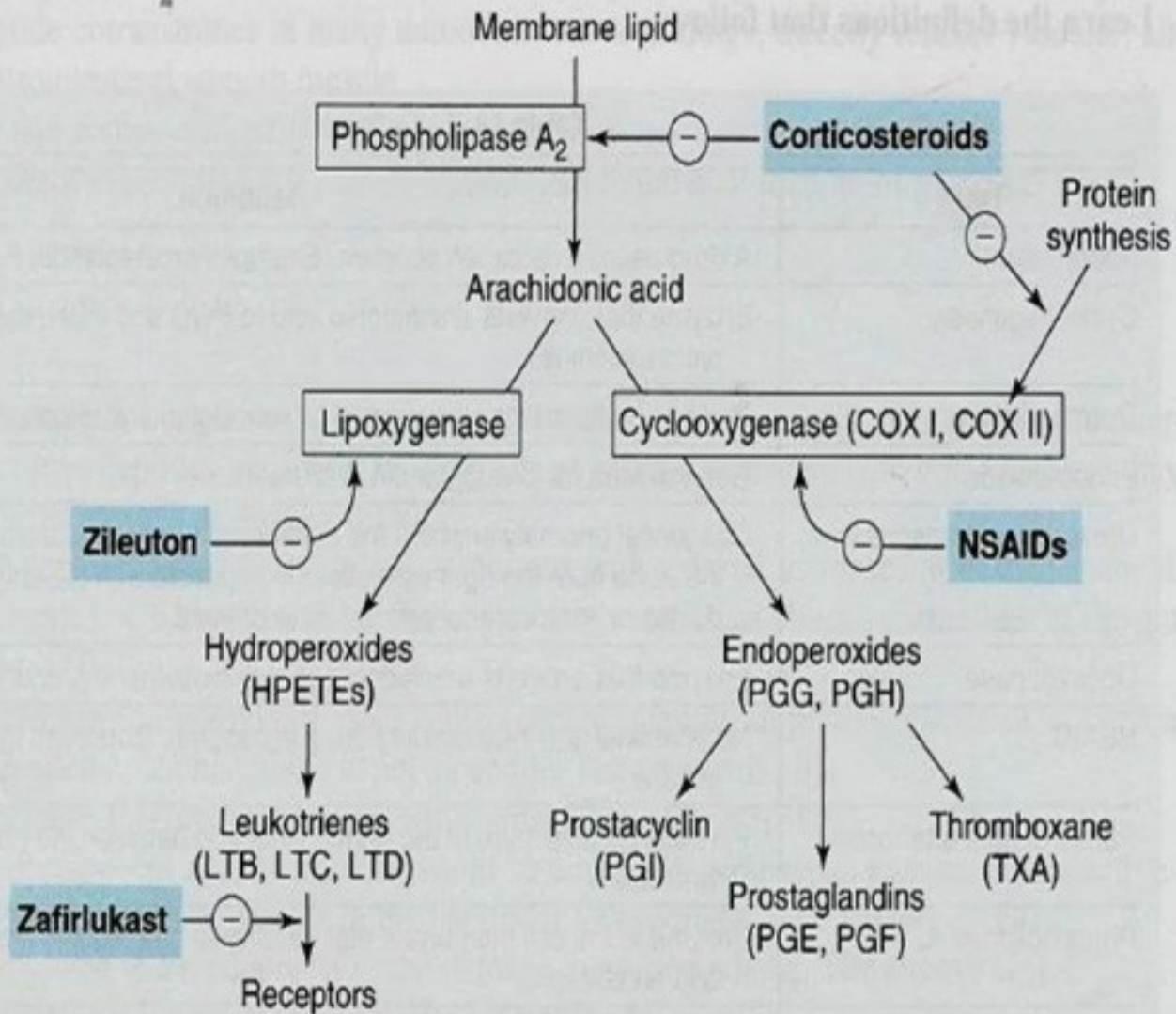
Side effects

- Bitter taste
- minor upper respiratory tract irritation (*burning sensation, nasal congestion*)

Leukotrienes antagonists

Leukotrienes

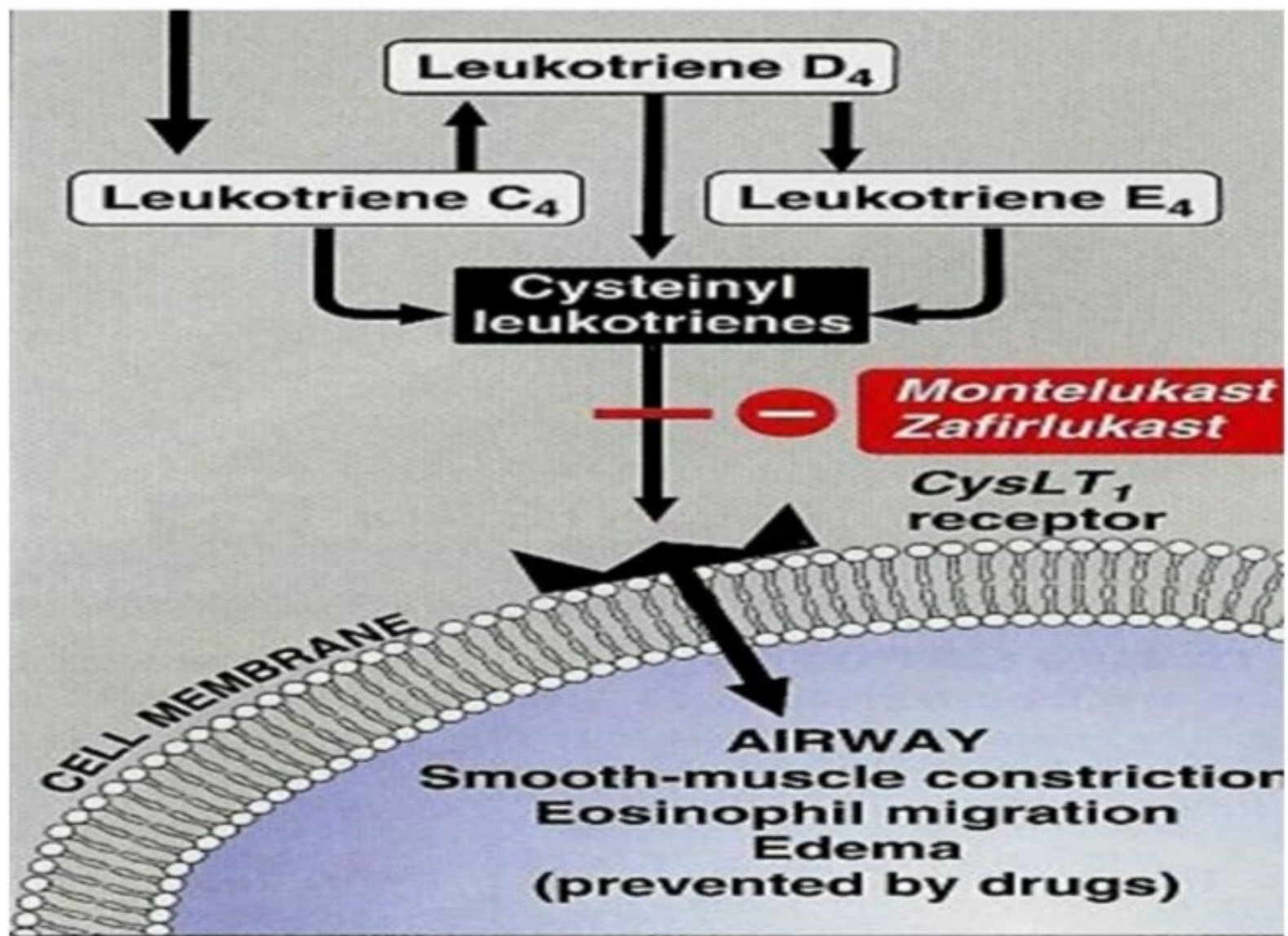
- **produced by the action of 5-lipoxygenase on arachidonic acid.**
- **Synthesized by inflammatory cells found in the airways (eosinophils, macrophages, mast cells).**
- **Leukotriene B4:** chemotaxis of neutrophils
- **Cysteinyl leukotrienes C4, D4 & E4:**
 - bronchoconstriction
 - increase bronchial hyper-reactivity
 - mucosal edema, mucus hyper-secretion



Leukotriene receptor antagonists

e.g. zafirlukast, montelukast, pranlukast

- are selective, reversible antagonists of cysteinyl leukotriene receptors (**CysLT₁receptors**).
- Taken orally.
- Are bronchodilators
- Have anti-inflammatory action
- Less effective than inhaled corticosteroids
- Have glucocorticoids sparing effect (potentiate corticosteroid actions).



Uses of leukotriene receptor antagonists

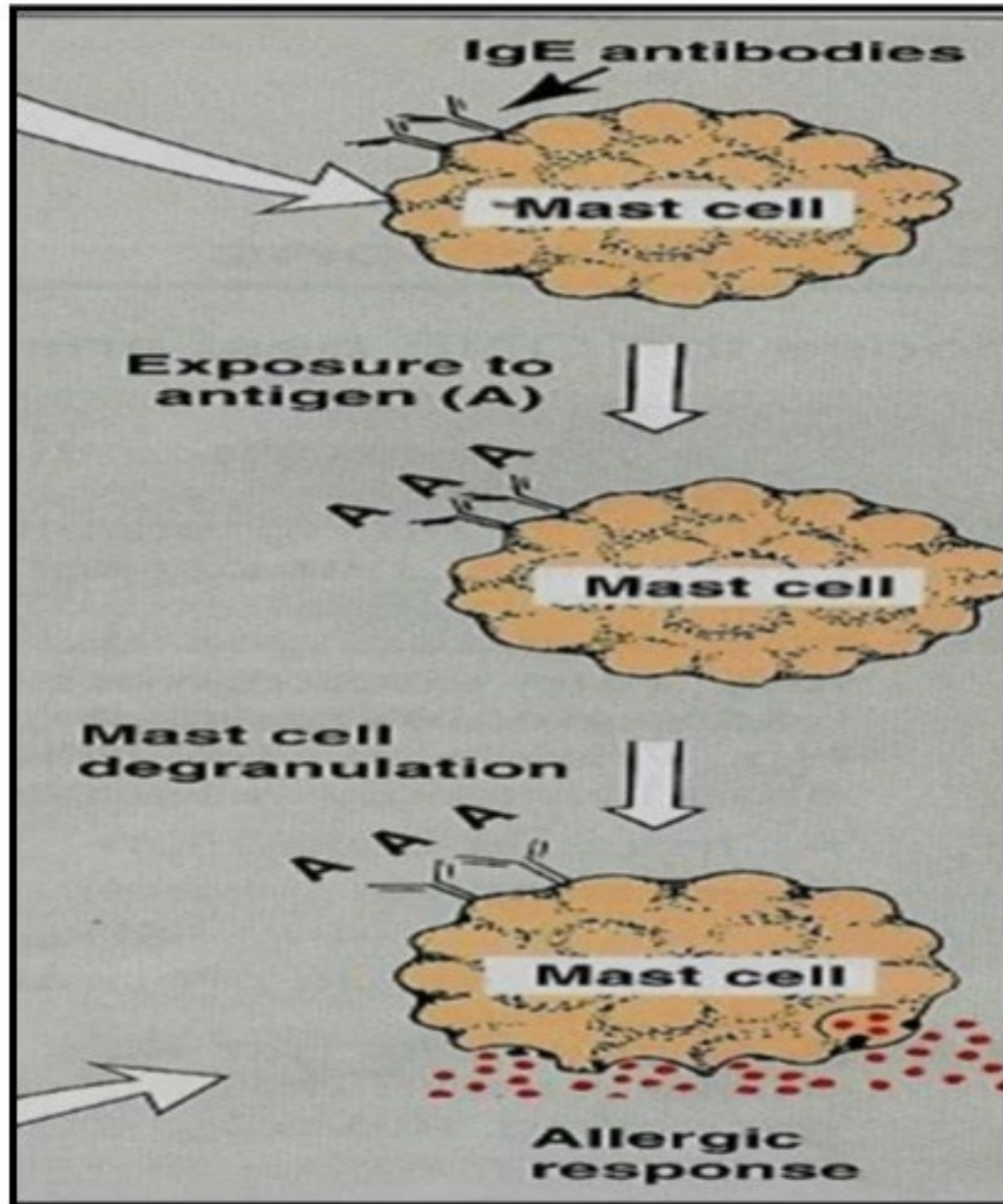
- Are not effective to relieve acute attack of asthma.
- **Prophylaxis** of mild to moderate asthma.
- Aspirin-induced asthma
- Antigen and exercise-induced asthma
- Can be combined with glucocorticoids (*additive effects, low dose of glucocorticoids can be used*).

Side effects:

Elevation of liver enzymes, headache, dyspepsia

Omalizumab

- is a monoclonal antibody directed against **human IgE**.
- prevents IgE binding with its receptors on mast cells & basophiles.
- ↓ release of allergic mediators.
- used for treatment of allergic asthma.
- Expensive-not first line therapy.



Status asthmatics

- Very sever and **sustained attack** of **asthma** which fails to respond to treatment with usual measures
- **Management**
 - Administration of oxygen
 - continuous administration of aerosolized **β2 agonis**
 - **Aminophylline** IV infusion
 - Iv fluid to avoid dehydration
 - **Antibiotics** in the presence of evidence of infection

Drugs used in COPD

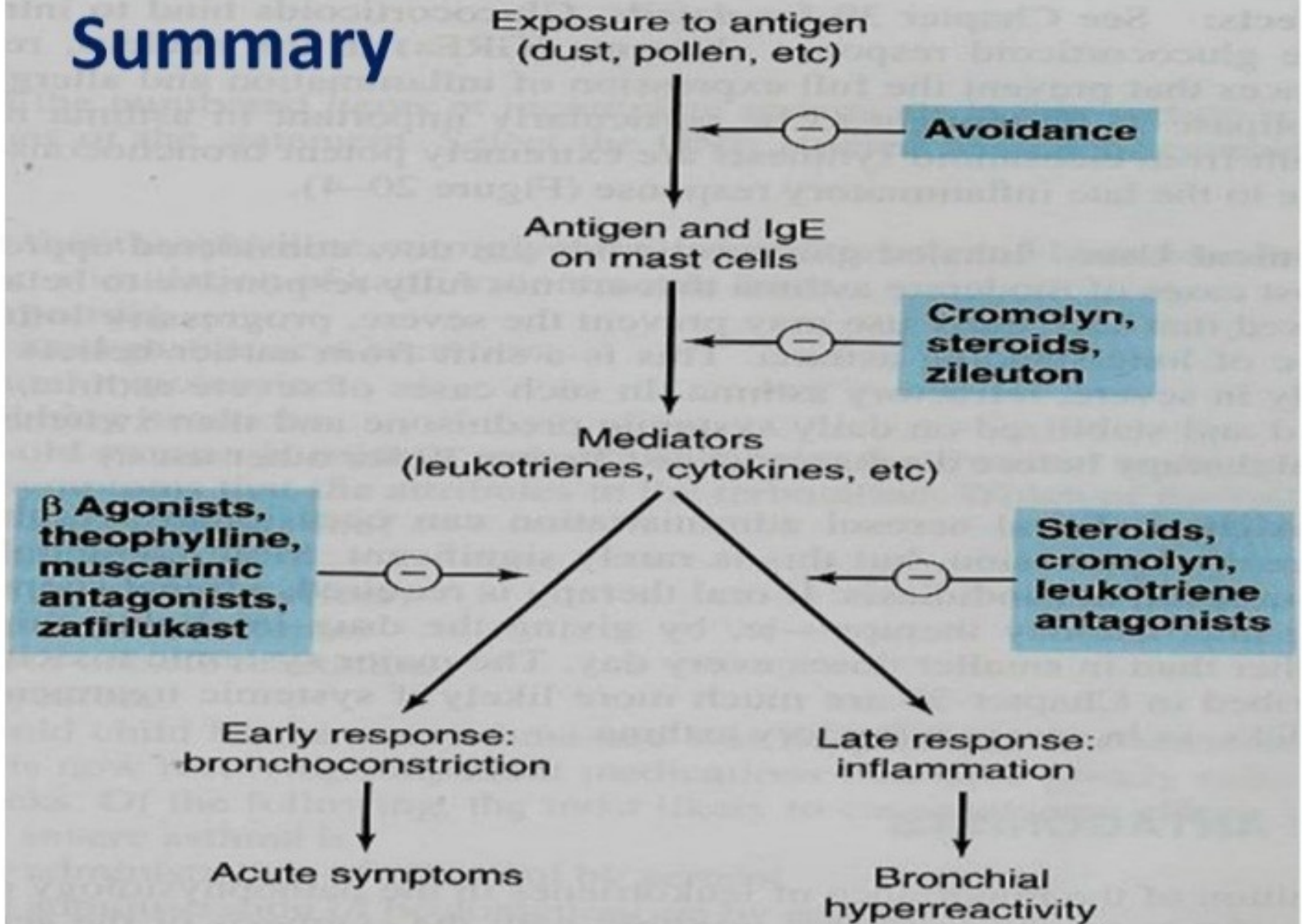
- COPD is a chronic irreversible airflow obstruction, lung damage and inflammation of the air sacs (alveoli).
- Smoking is a high risk factor
- Treatment:
 - Inhaled bronchodilators
 - Inhaled glucocorticoids
 - Oxygen therapy

Treatment of COPD

Inhaled bronchodilators

- Inhaled antimuscarinics **(are superior to $\beta 2$ agonists in COPD)**
- $\beta 2$ agonists
- these drugs can be used either alone or combined
 - salbutamol + ipratropium
 - salmeterol + Tiotropium *(long acting-less dose frequency).*

Summary



Bronchodilators (relievers for bronchospasm)

Drugs		
B2 agonists Salbutamol, terbutaline	– Short acting – main choice in acute attack of asthma – Inhalation	– Adenyl cyclase – cAMP
Salmeterol, formoterol	Long acting, Prophylaxis Nocturnal asthma	
Antimuscarinics Ipratropium (Short) Tiotropium (long)	Main drugs For COPD Inhalation Inhalation	Blocks M receptors
Xanthine derivatives Theophylline Aminophylline	 (orally) (parenterally)	•Inhibits phosphodiesterase↑ cAMP

Anti-inflammatory drugs (prophylactic)

Corticosteroids (Inhibits phospholipase A2) Dexamethasone, Fluticasone, budesonide	Inhalation
prednisolone	Orally
Hydrocortisone	parenterally
Mast stabilizers Cromoglycate (Cromolyn), Nedocromil	Inhalation, prophylaxis in children
Cysteinyl antagonists (CyLT1 antagoist) Zafirlukast	orally
Omalizumab (Anti IgE antibody)	Injection, SC

Anti-tussives

- ❖ Cough is a **protective reflex**, serves as
 - expelling **sputum** and **other irritants** from **respiratory airway**.
- ❖ Cough can be classified as
- ❖ Useful productive cough
 - Effectively expels secretions and exudates
- ❖ Useless cough
 - Non-productive chronic cough
 - Can be due to smoking and local irritants
- ❖ **Anti-tussives** are drugs used to suppress the **intensity** and **frequency of coughing**.

Anti-tussives...

❖ Central anti- tussives

- ✓ Suppress the medullary cough center and can be
 - Opioid antitussive e.g. codeine, hydrocodeine
 - Non opioid antitussives e.g. dextromethorphan

❖ Peripheral antitussives

- ✓ Decrease the input of stimuli from the cough receptor in the **respiratory passage**
E.g Demulcents e.g. liquorices lozenges, honey

Codeine

- ❖ It is a **narcotic relatively less addicting** drug and central antitussive agent
- ❖ **side effects**
 - dryness of mouth,
 - constipation and
 - dependence.

Dextrometorphane

- It is opoid synthetic **antitussive**, essentially **free of analgesic and addictive properties**
- **side effect....** respiratory depression

Expectorant: is a drug that aid in removing **thick persistent mucus** from respiratory passage

E.g: sodium citrate, saline expectorant

Mucolytics are agents that liquefy mucus and facilitate expectoration, e.g. Ambroxol, acetylcysteine

Decongestants

❖ Decongestants are the drugs that reduce congestion of **nasal passages**, which in turn open clogged **nasal passages** and enhances **drainages of the sinuses**

e.g phenylephrine, oxymetazoline etc

Mechanisms of action

- They are $\alpha 1$ agonists, which produce localized vasoconstriction on the small blood vessels of the nasal membrane.
- Reduce congestion in nasal passages

Decongestants...

- ❖ Used in congestion associated with rhinitis, hay fever, allergic rhinitis and to a lesser extent common cold
- ❖ Drugs can be administered nasally or orally for longer duration of action

Classifications of nasal decongestant

Short acting decongestants administered topically –
phenylephrine, phenylpropanolamine

Long acting decongestants administered orally
ephedrine, pseudoephedrine, naphazoline

Long acting topical decongestants

- Xylometazoline
- Oxymetazoline